

Ex. No: 12 Named Entity Recognition System

Implement Named Entity Recognition System for extracting Named Entities such as People, Organizations and Locations over any small and large corpus datasets

Datasets:

- 1. Any Small Corpus
- 2. Any Large Corpus

Aim

To implement a Named Entity Recognition (NER) system that automatically identifies and classifies named entities — People (PER), Organizations (ORG), and Locations (LOC) — in text from small and large corpus datasets.

Procedure/Pseudocode

- 1. Load a NER-annotated corpus such as CoNLL-2003 (English) for training and evaluation.
- 2. Implement a rule-based NER baseline using gazetteer lists (known person names, country/city names, organization names) and pattern rules (capitalized words, title patterns).
- 3. Extract features for each token: word, prefix/suffix, POS tag, capitalization, previous/next word context, and gazetteer membership.
- 4. Train a CRF (Conditional Random Field) model using sklearn-crfsuite on IOB-tagged training data.
- 5. Implement a Bi-LSTM-CRF neural NER model using PyTorch for improved accuracy on large corpora.
- 6. Apply the model to extract entities from test corpus; decode using Viterbi for CRF or greedy decoding for neural model.
- 7. Post-process: merge consecutive entity tokens using IOB2 scheme (B-PER, I-PER, O).
- 8. Evaluate using precision, recall, and F1 score per entity class and overall.

Libraries Used

- 1. spacy – for pre-trained NER and tokenization
- 2. sklearn-crfsuite – for CRF-based NER
- 3. torch – for Bi-LSTM-CRF neural NER
- 4. nltk – for rule-based chunking and gazetteers
- 5. seqeval – for NER evaluation metrics (entity-level F1)

Test Cases

- 1. Input: 'Barack Obama visited Google headquarters in Mountain View.' → PER: Barack Obama, ORG: Google, LOC: Mountain View
- 2. Input: 'Apple Inc. announced its new product in Cupertino, California.' → ORG: Apple Inc., LOC: Cupertino, LOC: California
- 3. IOB encoding: 'New B-LOC York I-LOC City I-LOC' → Entity: New York City (LOC)
- 4. Ambiguity test: 'Amazon' → ORG (tech company) vs LOC (river) — resolved by context

Results

The CRF-based NER system achieved an overall F1 score of approximately 85% on the CoNLL-2003 test set, with Person entities achieving highest F1 (88%) and Location entities achieving 84%. The Bi-LSTM-CRF neural model improved overall F1 to 91%. SpaCy's pre-trained model achieved 86% F1 out of the box on the test corpus.